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1.5 cost and benefit analysis

1. After graduating from high school, Maria chose to go to college, while Omar chose to work full- time. Which of the following best describes the opportunity costs for these decisions?

(A) Maria's opportunity cost includes the salary she could have earned if she had gone to work. ✓

(B) Maria's opportunity cost is her living expenses while attending college.

(C) Omar's opportunity cost is the salary he will earn from working.

(D) Omar's opportunity cost is the tuition and expenses he would have paid for college.

(E) Omar's opportunity cost is definitely greater than Maria's.

2. After graduating from high school, Peggy Smith decided to enroll in a two-year program at the local community college rather than to accept a job that offered a salary of $12,000 per year. If the annual tuition and fees are $4,600, the annual opportunity cost of attending the community college is

(A) $4,600

(B) $7,400

(C) $12,000

(D) $16,600 ✓

(E) $24,000

3. All of the following are included in computing the opportunity cost of attending college EXCEPT

(A) interest paid on student loans

(B) wages the student gave up to attend college

(C) money spent on college tuition

(D) money spent on clothing expenses ✓

(E) money spent on books and supplies

4. An opportunity cost is entailed in which of the following situations?

I. A student decides to attend college full-time.

II. A family uses its $20,000 savings to purchase an automobile.

III. A farmer decides to grow more wheat and less corn.

(A) I only

(B) II only

(C) III only

(D) I and III only

(E) I, II, and III ✓

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5. A city is considering building playgrounds in city parks located near residential areas. The city planner presented to the township the following cost and benefit estimates to help the city council decide how many playgrounds to build.

Number of Playgrounds	Total Cost ($)	Total Benefit ($)
1	20,000	30,000
2	50,000	70,000
3	80,000	105,000
4	110,000	140,000
5	160,000	150,000
6	170,000	165,000

What is the optimal number of playgrounds to maximize net benefits?

(A) 6

(B) 5

(C) 4 ✓

(D) 3

(E) 2

Answer C

Correct. Building 4 playgrounds maximizes net benefits
(net benefits = total benefits - total costs = $140,000-$110,000 = $30,000).

6. Using cost-benefit analysis, a local government would decide to build a new bridge if

(A) The additional tax paid by an individual resident is greater than the additional benefit of building the bridge for all residents.

(B) The toll paid by an individual resident crossing the bridge is less than the resident's benefit from crossing the bridge.

(C) The total costs of building the bridge are less than the total benefits from building the bridge. ✓

(D) The total costs of building the bridge are greater than the total benefits from building the bridge.

(E) Total costs are at a minimum and total benefits are at a maximum.

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Answer C

Correct. Cost-benefit analysis involves comparing total costs and total benefits to maximize net benefits.

7.

Hours at the Library	Total Benefit	Total Cost
1	12	10
2	20	13
3	26	18
4	31	25
5	33	33
6	35	42

The table provided shows the total benefits and costs of studying at the library. If the student is rational, the optimal number of hours to spend at the library is

(A) 2

(B) 3 ✓

(C) 4

(D) 5

(E) 6

Answer B

Correct. The optimal number of hours to spend at the library is achieved when the marginal benefit is greater than or equal to the marginal cost. When the student spends 3 hours, the marginal benefit (6) is greater than the marginal cost (5), therefore the student should spend more hours studying in the library as long as the marginal benefit of the next hour exceeds the marginal cost. If the student spends the next hour, the fourth hour, at the library, the marginal cost (7) will exceed the marginal benefit (5). Thus, the student should spend three hours and not spend the fourth hour studying at the library.

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8. The opportunity cost of owning a business is equal to which of the following?

I. The economic profits earned in the business

II. The accounting profits earned in the business

III. The profits that could be earned in another business using the same amount of resources

(A) I only
(B) II only
(C) III only ✓
(D) I and III only
(E) I, II, and III

9. Which of the following best defines opportunity cost?

(A) It is the cost of producing those goods most desired by a given economy.
(B) It is the cost of the input mix that will lead to the greatest rate of growth for a given company.
(C) It is the amount of one product that must be given up in order to produce an additional unit of another product. ✓
(D) It is the use of the least-cost method of production.
(E) It is the cost of labor used in the production process.

10. Karen works part-time at a local convenience store and earns $10 per hour. She wants to spend next Saturday afternoon attending a music concert. The full price of a concert ticket is $75, but Karen was able to get a discounted price of $50 from a friend who purchased the ticket but has become unable to attend. If Karen took 4 hours off from her job to attend the concert, what was her opportunity cost of attending the concert?

(A) $40
(B) $50
(C) $75
(D) $90 ✓
(E) $115

11. Mr. Carpenter devotes his working time to producing tables and chairs. An increase in the demand for chairs will result in

(A) an increase in the amount of time he devotes to producing tables
(B) an increase in his opportunity cost of producing tables ✓
(C) a decrease in the price of lumber
(D) a decrease in the price of chairs
(E) a decrease in his total revenue

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12. Oren's father tells Oren he can have one dessert after dinner. He can choose from a scoop of ice cream, a slice of apple pie, a cup of chocolate pudding, or a piece of fruit. Oren prefers chocolate pudding to a piece of fruit; he prefers apple pie to chocolate pudding; and he prefers ice cream to apple pie. If Oren chooses a scoop of ice cream, what is his opportunity cost?

(A) A piece of fruit.

(B) A cup of chocolate pudding.

(C) A slice of apple pie. ✓

(D) A scoop of ice cream.

(E) A piece of fruit, a cup of pudding and a slice of apple pie.

Answer C

Correct. Apple pie is the next best alternative to ice cream. By choosing the ice cream, Oren gave up the opportunity to enjoy apple pie, the next best alternative.

13. Opportunity cost is defined as

(A) excess demand for a good

(B) unlimited resources and unlimited wants

(C) a comparative advantage in the production of a good

(D) the value of the next best alternative that is forgone when an activity is pursued ✓

(E) the value of all other possible alternatives that are forgone when an activity is pursued

14. Cost-benefit analysis assumes rational agents do which of the following?

(A) Always make a decision that is best for society

(B) Compare additional costs and additional benefits when making a decision ✓

(C) Consider only what they will lose when making a decision

(D) Consider only what they will gain when making a decision

(E) Always make the same decision

Answer B

Correct. To maximize benefit, rational individuals compare additional costs and additional benefits when making a decision. As long as the additional benefit of an action exceeds the additional cost it will be beneficial to undertake the activity.

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15. The amount of product Z that must be forgone in order to obtain some amount of product Y is called

(A) factor payments

(B) opportunity cost ✓

(C) marginal product

(D) income effect

(E) substitution effect

16. Assume a worker can either cut grass or wash windows in a given day. It takes 5 hours to mow a one-acre lawn and 2.5 hours to wash all the windows in a house. When a worker chooses to wash the windows of four houses in a neighborhood, what is the opportunity cost of this decision?

(A) The pay the worker receives for washing the windows in the four houses

(B) The pay the worker would receive for cutting one lawn

(C) Cutting the grass at the same four houses

(D) The pay the worker would have received from cutting the grass at two houses ✓

(E) The pay the worker would receive for cutting .25 lawns

Answer D

Correct. The opportunity cost of washing windows is the next-best alternative the worker has given up to wash windows. If it takes 5 hours to mow the one-acre lawn and 2.5 hours to wash the windows, the opportunity cost of mowing one lawn is washing the windows of two houses; the opportunity cost of washing the windows of one house is mowing .5 lawns. Thus, in washing the windows of four houses the worker is expending 10 hours, the time in which he could cut two lawns; or 4 houses multiplied by .5 lawns equals two lawns and the pay that he would have earned.

17.

Project	Total Costs(in millions)	Total Benefits(in millions)
Widening existing highways	$100	$80
Building new highways	$100	$90
Building a new bridge	$100	$140
Building a new tunnel	$100	$115
Expanding an existing airport	$100	$130

The table above shows the costs and benefits for new construction projects the government is considering. Based on these costs and benefits, which of the construction projects should the government undertake?

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(A) Widening existing highways
(B) Building new highways
(C) Building a new bridge ✓
(D) Building a new tunnel
(E) Expanding an existing airport

Answer C

Correct. Cost-benefit analysis involves comparing total costs and total benefits to maximize net benefits. When comparing the net benefits of each choice, net benefits are maximized when building a new bridge ($40 million).